FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 16-09-2020

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 4, 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

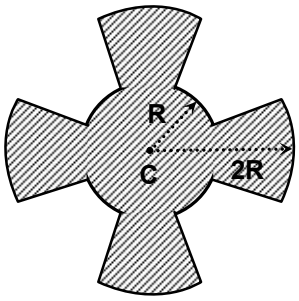
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A non-conducting portion of circular disc (as shown in figure) having positive charge density σ. Each cross section subtends an angle 45° on its center 'C'. A positive point charge q is slowly brought from infinity to the centre 'C'. Choose the correct option(s).



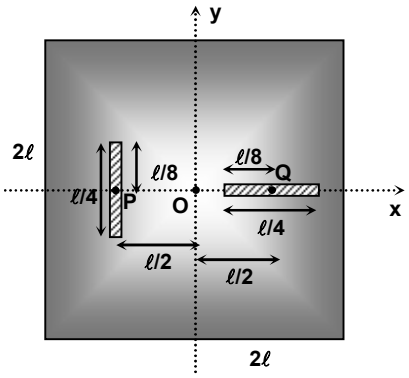
(A) The work done by the external agent is positive.

(B) The potential due to the non conducting portion at a distance of $\sqrt{3}R$ on its axis from centre C is $\frac{\sqrt{7}}{4}\frac{\sigma R}{\varepsilon_0}$.

(C) The potential due to the non conducting portion at a distance of $\sqrt{3}R$ on its axis from centre C is $\frac{\sigma R}{4\varepsilon_0}\left[2+\sqrt{7}-2\sqrt{3}\right]$.

(D) The magnitude of work done by the external agent to bring the charge particle slowly from infinity to the centre is $\frac{3}{4}\frac{\sigma Rq}{\varepsilon_0}$.

2. Two rods each of mass 'm' and length 'ℓ/4' are hinged on a larger square plate of same mass and edge length 2ℓ. Both rods can freely rotate about hinge points 'P' and 'Q' in the plane of square plate. Choose the correct option(s).



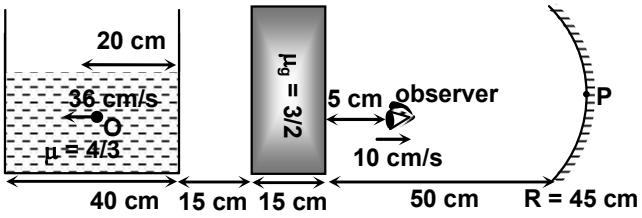
(A) The moment of inertia of the system about x-axis is $\frac{65}{192}m\ell^2$

(B) The moment of inertia of the system about y-axis is $\frac{161}{192}m\ell^2$

(C) The moment of inertia of the system about z-axis is $\frac{226}{192}m\ell^2$

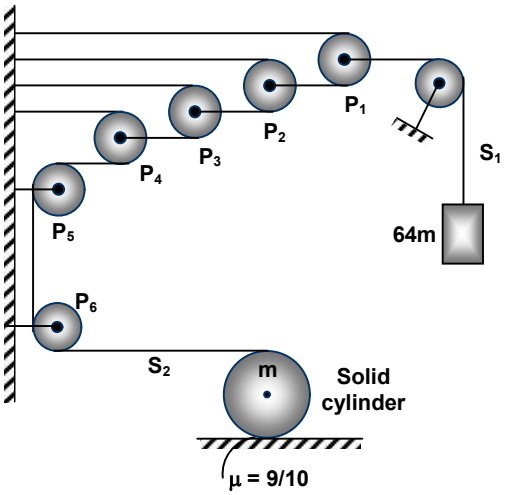
(D) The moment of inertia of the system about z-axis is $\frac{224}{192}m\ell^2$

3. A point object 'O' lies inside a transparent cuboidal container (of negligible thickness) having water as shown in the figure. An observer is situated between the glass slab and the concave mirror. At t = 0 observer starts moving with a velocity 10 cm/s towards right side and object 'O' also starts moving towards left side with a velocity 36 cm/s on the optic axis of concave mirror. Then choose the correct option(s).



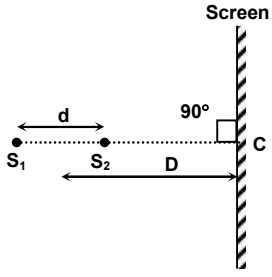
(A) Distance of image formed by the mirror from its pole is 30 cm left.
(B) Distance of object 'O' as seen by observer from the observer is 45 cm.
(C) The magnitude of velocity of the object 'O' with respect to observer when the observer directly sees the object is 37 cm/s.
(D) The magnitude of velocity of the image of object 'O' formed by the concave mirror with respect to observer is 13 cm/s.

4. All the pulleys and strings are massless. One end of the string S_1 is connected with a block and one end of the string S_2 is connected with a solid cylinder as shown in the figure. The coefficient of friction between the cylinder and the horizontal surface is $\mu = \frac{9}{10}$. Choose the correct option(s). (Take m = 2kg and g = 10 m/s^2)



(A) The tension in the string attached with the block '64m' is 48 Newton.
(B) The acceleration of block 64m is 4 m/s^2 downward.
(C) The acceleration of centre of mass of the solid cylinder is 32 m/s^2.
(D) Friction force between the cylinder and the horizontal surface is 16 N.

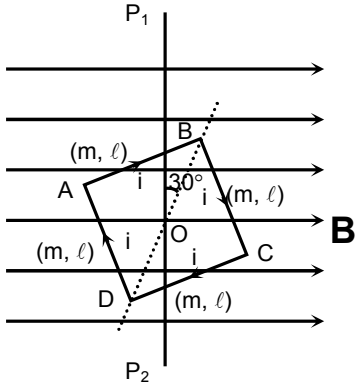
5. Two coherent sources are emitting light of wavelength λ = 5000Å. The separation between the sources is 2.5mm and the separation between the source and the screen is 2m. Then which of the following are correct.



(A) The position of 3000th order bright fringe from C is $\frac{8}{3}$ m.
(B) The position of 2500th order bright fringe from C is $2\sqrt{3}$ m.
(C) The position of 5th bright fringe from C is 0.08 m (include the bright fringe at C).
(D) The shape of fringes on the screen are circular.

6. In the α-decay of a U-238 nucleus the energy released in the decay is Q. The U-238 nucleus was initially stationary. Then which of the following is/are correct.
   (A) Ratio of kinetic energy of α-particle and thorium nucleus is 117 : 2
   (B) Ratio of momentum of α particle and thorium nucleus is 2 : 117
   (C) Momentum of α-particle is $\left(\frac{234Qm_\alpha}{119}\right)^{1/2}$
   (D) Recoil velocity of thorium nucleus is $\left(\frac{234Q}{13923\, m_{Th}}\right)^{1/2}$

7. Rod AB, BC, CD & DA for a square loop having current i, mass and length of each rod is m and ℓ respectively, is situated in a uniform magnetic field B which is parallel to the plane of paper as shown in the figure and it can rotate about fixed non conducting axis P_1P_2 perpendicular to $\vec{B}$ and rod AB, rod CD are attached with the axis, which lies in the plane of paper. At t = 0 the angle between the plane of the loop and the magnetic field is 30° and point C is above the plane of the paper (O is geometrical center of the loop). Then choose the correct option(s).



   (A) Angular acceleration of the square loop at t = 0 is $\frac{3\sqrt{3}}{4}\frac{Bi}{m}$.
   (B) Angular acceleration of the square loop at t = 0 is $\frac{3}{4}\frac{Bi}{m}$.
   (C) Angular velocity when square loop rotated by 60°, is $\sqrt{\frac{3Bi}{2m}}$.
   (D) Angular velocity when square loop rotated by 60°, is $\sqrt{\frac{\sqrt{3}Bi}{m}}$.

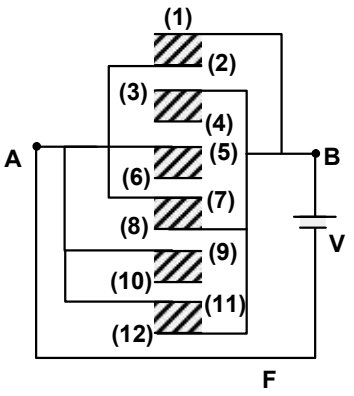
8. A uniform cylinder made of material of density ρ = 125 kg/m^3 pressed down and held at rest in a pool so that its upper circular surface is in level with the water surface. Length of the cylinder is ℓ = 5.40 m and radius is r = 10 cm. When released the cylinder jumps out of water. Water is calm while releasing, the axis of the cylinder is vertical, the cylinder comes out moving vertically. Then choose the correct option(s). (density of water is ρ_0 = 1000 kg/m^3 and acceleration due to gravity is g = 10 m/s^2)
   (A) The magnitude of velocity when the cylinder just leaves the water surface is 18 m/s.
   (B) The maximum height above the initial position attained by the cylinder after releasing is 16.2 m.
   (C) The maximum height above the initial position attained by the cylinder after releasing is 21.6 m.
   (D) The loss in potential energy of liquid when the cylinder comes out from the liquid, is 1458π Joule.

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

All identical capacitor plates each of area A are arranged such that adjacent plates are at d distance apart. Plates are connected to a source of emf V volts as shown in figure (dielectric constant in each shaded region is k = 2). Then



	List – I		List – II
(I)	Charge on plate 3 is $\frac{10\varepsilon_0 AV}{Kd}$. Then the value of K is	(P)	zero
(II)	The heat rejected by the system is $\frac{26\varepsilon_0 AV^2}{Kd}$. Then the value of K is	(Q)	3
(III)	Equivalent capacitance between AB is $\frac{26A\varepsilon_0}{Kd}$. Then the value of K is	(R)	$\frac{13}{5}$
(IV)	Potential difference between the plates 4 and 7 is $\frac{KV}{5}$. Then the value of K is	(S)	$\frac{26}{5}$
		(T)	6
		(U)	$\frac{10}{3}$

9. Which of the following is correct?

Options

(A) $I \rightarrow U, II \rightarrow Q, III \rightarrow P, IV \rightarrow Q$

(B) $I \rightarrow T, II \rightarrow P, III \rightarrow Q, IV \rightarrow U$

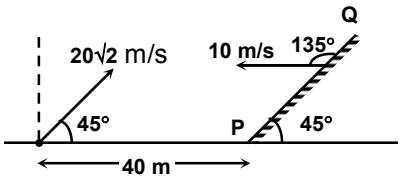
(C) $I \rightarrow U, II \rightarrow T, III \rightarrow Q, IV \rightarrow P$

(D) $I \rightarrow T, II \rightarrow T, III \rightarrow Q, IV \rightarrow U$

10. If the plates 4 and 5 are connected by a conducting wire. Then which of the following is correct.
Options
(A) I $\rightarrow$ U, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ P
(B) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ S
(C) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ U, IV $\rightarrow$ P
(D) I $\rightarrow$ P, II $\rightarrow$ U, III $\rightarrow$ R, IV $\rightarrow$ P

Answer the following by appropriately matching the list based on the information given in the paragraph.

A ball is projected with a velocity $20\sqrt{2}$ m/s at an angle of 45° with the horizontal as shown in the figure and a very large wall is also moving with a velocity 10 m/s horizontally maintaining its inclination 45° with the horizontal and the initial separation is 40m as shown in the figure . {Take $\tan^{-1}(2) = 63°$, $\tan^{-1}(1/3) = 18°$ and g = 10 m/s^2}



	List – I		List – II (in S.I. unit)
(I)	Time after which the ball collides with the wall is	(P)	8
(II)	Magnitude of velocity of the ball after the collision is	(Q)	2
(III)	Maximum height attained by the ball from the ground is	(R)	$10\sqrt{10}$
(IV)	After the collision, ball moves at an angle of $(2.25 \times k)°$ with the inclined wall as observed by the observer which is fixed on the moving wall, then k is	(S)	40
		(T)	20
		(U)	65

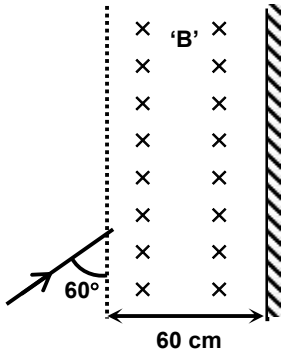
11. If the collision between the wall and the ball is perfectly elastic. Then which of the following is correct?
Options
(A) I $\rightarrow$ Q, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ Q
(B) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ U
(C) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ U, IV $\rightarrow$ T
(D) I $\rightarrow$ T, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ U

12. If coefficient of restitution between the ball and the wall is $\frac{1}{3}$. Then which of the following is correct?
Options
(A) I $\rightarrow$ U, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ P
(B) I $\rightarrow$ Q, II $\rightarrow$ T, III $\rightarrow$ S, IV $\rightarrow$ P
(C) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ U, IV $\rightarrow$ P
(D) I $\rightarrow$ P, II $\rightarrow$ U, III $\rightarrow$ R, IV $\rightarrow$ P

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A positive charged particle q and mass m enters and comes out from a uniform magnetic field B which exists with direction inside the paper in a finite space of width 60 cm after collision with the wall (collision is perfectly elastic) as shown in the figure. If time elapsed by the charge particle inside the magnetic field is $\left(\frac{23^\circ}{90^\circ}\right)\frac{\pi m}{qB}$. The radius of curvature of the charge particle is $\left(\frac{1}{2k}\right)$ meter when it is inside the magnetic field. Find the value of k.

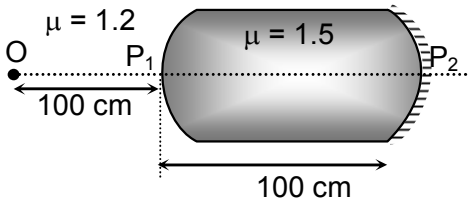


14. Find the position of centre of mass (in meter) of the quarter solid sphere from its geometrical centre in which mass per unit volume is given as $\rho(r)=\rho_0\left(1-\frac{r^2}{R^2}\right)$, where r is radial distance from geometrical centre and R(= 8 m) is radius of solid quarter sphere. (Take $\sqrt{2}=1.4$ and $\sqrt{3}=1.7$)

15. In a PO box experiment it is found that for P : Q = 100 : 10, the deflection is to the left by 4 division when R is 545 Ω and to the right by 6 division when R is 544 Ω. Find the correct value of the unknown resistance (in Ω) in the fourth arm. (Assume that the deflection changes linearly with R near the balance point)

16. A charged particle is moving with a velocity $\vec{v}=-\hat{i}+\hat{j}-\hat{k}$ and the electric field produced by it at a distant point is $\vec{E}=\hat{i}+\hat{j}+\hat{k}$. The magnetic field produced at the same point due to the same charged particle is $K\mu_0\varepsilon_0$. Find the value of K. (All quantities are in S.I. units, take $\sqrt{2}=1.4$ and $\sqrt{3}=1.7$)

17. A luminous point object O is placed in front of spherical glass surface of radius R = 20 cm. The final real image is formed after refraction through the spherical surface, followed by reflection at the concave mirror of radius R = 400 cm and a second refraction through the spherical surface. Find the distance (in centimeter) of this final image from the pole P_1 of the spherical surface.



18. The main scale of a vernier calipers reads in millimeter and its vernier is divided into ten divisions which coincide with eight division of main scale. Further more when a cylinder is tightly placed along its length between two jaws, it is observed that the zero of vernier scale lies just right to 35th division of main scale and sixth division of vernier scale coincides with the main scale. Then find the measured value in centimeter. There is no zero error.

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. For the following process

$$\text{Ice} \underset{}{\overset{0^\circ C}{\rightleftharpoons}} H_2O(\ell)$$

Which of the following is/are true?

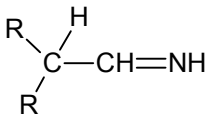
(A) $\Delta G > 0$

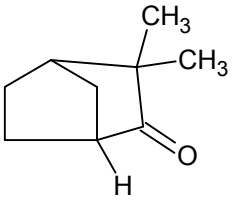
(B) $\Delta H > 0$

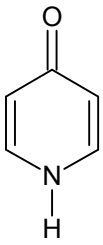
(C) $\Delta S > 0$

(D) $\Delta U > 0$

20. Tautomerism can be shown by the given molecule(s)

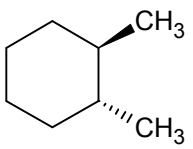
(A) 

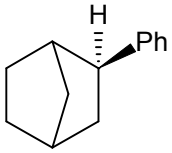
(B) 

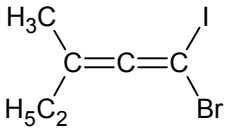
(C) 

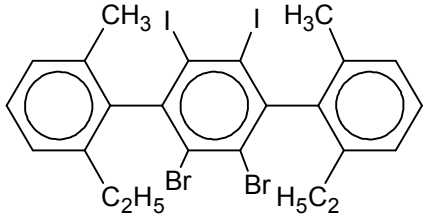
(D) CH_3CH_2NO

21. Optically active molecule(s) is/are

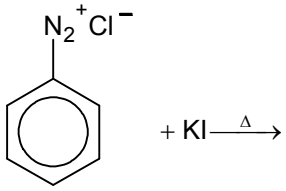
(A) 

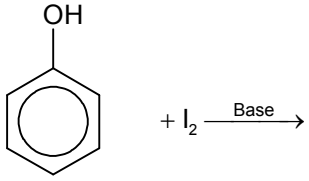
(B) 

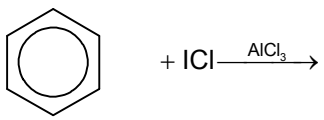
(C) 

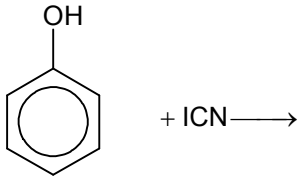
(D) 

22. Which of the following method(s) is/are correct to get iodination product of aryl group?

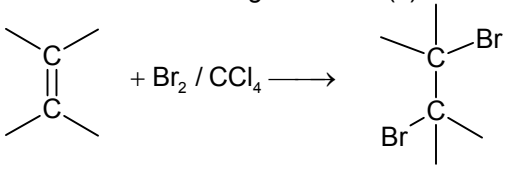
(A) 

(B) 

(C) 

(D) 

23. Which of the following statement(s) is/are correct for the given reaction?



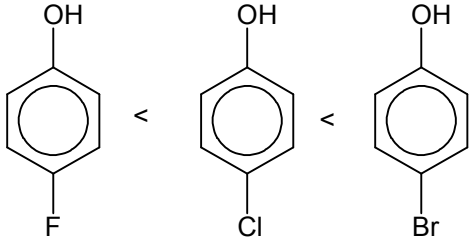
(A) Reaction will stop at high temperature.
(B) In reaction anti-addition of bromine takes place via ionic mechanism.
(C) Vacant d-orbital of Br_2 interacts with π bond of alkene
(D) Non-bonding electron pair of Br^- ion attacks on π^* orbital of alkene

24. Which of the following statement(s) is/are correct?

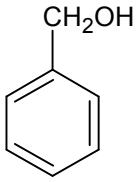
(A) order of rate of reaction for S_NAr mechanism in ArX is ArF > ArCl > ArBr > ArI

(B) Rate of reaction for benzyne mechanism in following Aryl Halides is ArBr > ArCl > ArF

(C) Acidic strength of substituted phenol are



(D) $CH_2 = CH - CHO \xrightarrow[\text{reagent}]{\text{Jone's}} CH_2 = CH - COOH$

25. 

$\xrightarrow{PCC} A \xrightarrow{NH_2OH} B \xrightarrow{H^+} C + D$

$C \xrightarrow[(ii)\ H_2O]{(i)\ Br_2/alc.KOH} E \xrightarrow[alc.\ KOH]{CHCl_3} F$ bad smelling compound.

$D \xrightarrow[hydrolysis]{H_3O^+} E + HCOOH$

Which of the following is/are correct?

(A) C is $Ph - \overset{O}{\overset{\|}{C}} - NH_2$

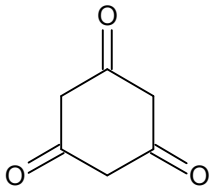
(B) D is $H - \overset{O}{\overset{\|}{C}} - NHPh$

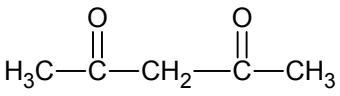
(C) E is $PhNH_2$

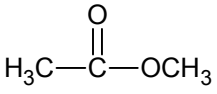
(D) F is Ph – NC

26. Haloform reaction is given by which of the following compound(s)

(A) $CH_3 - CH_2Cl$

(B) 

(C) 

(D) 

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

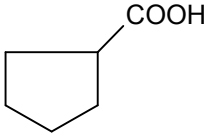
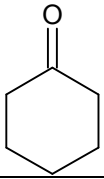
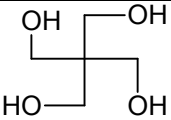
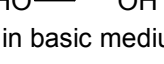
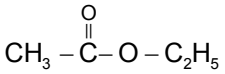
Answer the following by appropriately matching the list based on the information given in the paragraph.

List–I		List–II	
(I)	Pb^{2+}	(P)	White ppt. with KI
(II)	Ag^{+}	(Q)	Yellow ppt. with KI which dissolves in excess of KI
(III)	Cu^{2+}	(R)	Brick red ppt. with K_2CrO_4
(IV)	Bi^{3+}	(S)	Black ppt. with KI which dissolves in excess of KI to give orange solution
		(T)	Brown ppt. with $K_4[Fe(CN)_6]$
		(U)	Yellow ppt. with KCN which turns readily into white

27. Which of the following is correctly matched?
(A) (I), (Q)
(B) (I), (P)
(C) (III), (S)
(D) (III), (R)

28. Which of the following is not correct?
(A) (III), (U)
(B) (III), (T)
(C) (IV), (S)
(D) (II), (P)

Answer the following by appropriately matching the list based on the information given in the paragraph.

List–I		List–II	
(I)	CH_3CHO	(P)	On reaction with (i) X_2/H^+, (ii) KOH  (iii) H_3O^+, gives
(II)	PhCHO	(Q)	On acyloin condensation gives 'α' hydroxyl ketone
(III)		(R)	 Gives  with excess of HCHO in basic medium
(IV)		(S)	On reaction with (i) NH_2OH (ii) H^+ (iii) polymerization gives Nylon-6
		(T)	Gives benzoin condensation with KCN/alc. KOH
		(U)	Gives Claisen condensation

29. Which of the following is INCORRECT?
(A) (I), (R)
(B) (II), (T)
(C) (III), (P)
(D) (IV), (S)

30. Which of the following is INCORRECT?
(A) (II), (Q)
(B) (III), (S)
(C) (IV), (Q)
(D) (IV), (U)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. Total number of possible stereoisomers of the following co-ordinate compound
Mabcdef where - a, b, c, d, e and f are mono dentate ligands.

32. $K_4[Fe(CN)_6]$, which dissociates 80% in aqueous solution. Calculate van't Hoff factor of $K_4[Fe(CN)_6]$.

33. If we remove all the ions along one body diagonal in zinc blende structure. The fraction of tetrahedral void remains occupied is X, then the value of 10X is:

34. A mixture of carbonates of two bivalent metals contains 14.2% of one metal. If 3.2 gm of the mixture liberates 1.44 g CO_2, what would be the % by weight of the other metal.

35. CH_3COOH (75 ml, 0.1 M) is titrated against 0.1 M NaOH solution. Calculate pH after the addition of 80 ml of NaOH. [log 0.23 = 0.6382]

36. What is the bond order of C – O bond in CO_3^{2-} ion?

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Let $Z=\sum_{r=1}^{100}(2r+3)\left\{\sum_{k=1}^{8}\left(\sin\frac{2k\pi}{9}-i\cos\frac{2k\pi}{9}\right)\right\}^{r}$, then which of the following option(s) is incorrect?

(A) $|Z|=100\sqrt{2}$

(B) $\text{Arg}\left(Z^2\right)=\frac{7\pi}{4}$

(C) $|Z|=50\sqrt{2}$

(D) $\text{Arg}\left(Z^2\right)=\frac{3\pi}{2}$

38. Let $N = 2007^{2013} – 1974^{2013} – 1946^{2013} + 1913^{2013}$, then

(A) N is divisible by 61

(B) N is divisible by 2013

(C) N is divisible by 28

(D) all of these

39. Let a parabola touches the line y = 2x and 2y + x = 0 at P(5, 10) and Q(2, –1) respectively, then which of the following options is/are correct?

(A) equation of directrix is 9y + 7x = 0

(B) equation of tangent at vertex is 182x + 234y – 125 = 0

(C) focus is $\left(\frac{55}{26},\frac{15}{26}\right)$

(D) length of latus rectum is $\frac{250}{13\sqrt{130}}$

40. Let $f(x) = |x^2 – 5x + 6| + |x^2 – 4x + 3| – ||x – 3|(|x – 1| – |x – 2|)|$; $x \in R$, then f(x) has a

(A) local extremum at $x = \frac{7}{2}$

(B) local maximum at $x = \frac{3}{2}$

(C) local minimum at x = 2

(D) local maximum at x = 1

41. Let $\hat{x}$, $\hat{y}$ and $\hat{z}$ be unit vectors such that $\hat{x}+\hat{y}-\hat{z}=\vec{p}$, $\hat{x}\times\left(\hat{y}\times\hat{z}\right)=\vec{q}$ and $\left(\hat{x}\times\hat{y}\right)\times\hat{z}=\vec{r}$.

If $|\vec{p}|=\sqrt{3}$, $\vec{q}\cdot\hat{y}=\frac{3}{4}$ and $\vec{p}$ is perpendicular to $\vec{x}$, then which of the following option(s) is/are correct?

(A) $\vec{r}\cdot\hat{y}=\frac{3}{4}$

(B) $\vec{q}\cdot\hat{x}=0$

(C) $|\vec{q}|=\frac{3}{4}$

(D) $|\vec{r}| = \frac{3}{8}$

42. If the polynomial $x^4 + px^3 + qx^2 + rx - 48$ is divisible by the polynomials $x^3 - 2ax^2 + bx - 6$ and $x^3 + ax^2 - (3b + 1)x + 48$, then which of the following option is/are correct?
(A) $b - a = 8$
(B) $p = 2$
(C) $q = -7$
(D) $r = -82$

43. Let $L_1 = \lim_{x\to 0} \frac{3ax + \sin\left((a-8)x\right)}{\left(\sin^{-1}x\right)^3}$; L_1 is finite,

$L_2 = \lim_{x\to\infty} \frac{8x^5 - \left(\text{cosec}^{-1}x\right)^5}{\frac{4}{\pi}x^5 \sec^{-1}(kx-1) + \frac{k^3x^{10}}{\pi}\tan^{-1}\left(\frac{\pi}{x^5}\right) + \frac{5k^2x^5}{2}}$.

If L_2 is finite and equal to a, then which of the following option is/are correct?

(A) $\frac{L_1}{L_2} = 18$

(B) $\frac{L_1}{L_2} = 9$

(C) $k = -2$

(D) $a + k = 2$

44. Let α, β, γ be three real numbers satisfying $[\alpha \ \beta \ \gamma]\begin{bmatrix} 4 & 1 & 7 \\ -1 & 3 & 8 \\ 3 & -4 & -9 \end{bmatrix} = [4 \ 2 \ 10]$ ..... S. If the point P(α, β, γ) with reference to S, lies on a plane (π) containing the lines $L_1 : \frac{x+9}{1} = \frac{y+8}{-2} = \frac{z-3}{1}$ and $L_2 : \frac{x+9}{-4} = \frac{y+8}{3} = \frac{z-3}{0}$, then which of the following option(s) is/are correct?

(A) Distance of plane (π) from the point $\left(0, -11, 2\sqrt{2}\right)$ is 2 units

(B) Line $L : \frac{x+10}{3} = \frac{y+6}{1} = \frac{z-2}{1}$ completely lies in plane (π)

(C) $6\alpha + \beta + \gamma$ is equal to 6

(D) If plane (π) intersect the coordinate axes at points A, B and C respectively then volume of tetrahedron OABC, (O being origin) is $\frac{(44)^3}{360}$ cubic units

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

MM-I. Let z = x + iy be a complex number represented by the point P(x, y) in the argand plane. List-I contains information about the region S consisting of set of points satisfying a fixed condition. List-II contains information about possible arguments of complex number Z lying in region S

List-I		List-II	
(I)	$\mathrm{Re}\left((1+i)Z^2\right) > 0\,;\mathrm{Re}(Z) \geq 0$	(P)	$\frac{3\pi}{8} < \arg Z < \frac{7\pi}{8}$
(II)	$\mathrm{Re}\left((1+i)Z^2\right) > 0\,;\mathrm{Re}(Z) < 0$	(Q)	$-\frac{5\pi}{8} < \arg Z < \frac{-\pi}{8}$
(III)	$\mathrm{Im}\left((1+i)Z^2\right) > 0\,;\mathrm{Im}(Z) < 0$	(R)	$-\frac{3\pi}{8} < \arg Z < \frac{\pi}{8}$
(IV)	$\mathrm{Im}\left((1+i)Z^2\right) < 0\,;\mathrm{Im}(Z) \geq 0$	(S)	$\frac{\pi}{2} < \arg Z < \frac{3\pi}{4}$
		(T)	$-\frac{3\pi}{8} < \arg Z < \frac{5\pi}{8}$
		(U)	$-\frac{\pi}{8} < \arg Z < \frac{\pi}{8}$

45. Which of the following is the only CORRECT combination?
(A) (I), (R), (U)
(B) (II), (Q), (T)
(C) (I), (R), (S)
(D) (III), (R), (Q)

46. Which of the following is the only CORRECT combination?
(A) (I), (Q), (S)
(B) (II), (R), (T)
(C) (III), (S), (T)
(D) (IV), (P), (S)

Answer the following by appropriately matching the list based on the information given in the paragraph.

MM-II. Let $f(x)=\begin{matrix}[2x] & ; & -1\le x<1\\ |2x| & ; & 1\le x\le 2\end{matrix}$, $g(x)=\frac{\operatorname{cosec} x}{2}$; $x \in (0, 2\pi) - \{\pi\}$ (where [.] denotes the greatest integer function),

List-I: contains information about f(f(x)), f(g(x)), f(2g(x)), g(f(x))

List-II: contains information about the continuity and differentiability of ff(x), f(g(x)), f(2g(x)), gf(x)

List-I		List-II	
(I)	$f(g(x)) = -2 \ \forall \ x \in \left[\frac{5\pi}{4}, \frac{5\pi}{3}\right] - \left\{\frac{3\pi}{2}\right\}$	(P)	discontinuous at x = 1
(II)	g(f(x)) = 1 for some $x \in (1, 2)$	(Q)	continuous at $x = \frac{\pi}{2}$
(III)	$f(f(x)) = 0$ for $\forall \ x \in \left[\frac{1}{4}, \frac{1}{3}\right]$	(R)	discontinuous at $x = \frac{5\pi}{6}$
(IV)	$f(2g(x)) = 2$ for some $x \in \left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$	(S)	continuous at $x = \frac{\pi}{6}$
		(T)	differentiable at $x = \frac{\pi}{2}$
		(U)	differentiable at $x = \frac{\pi}{6}$

47. Which of the following is the only CORRECT combination?
(A) (I), (P), (T)
(B) (I), (R), (T)
(C) (II), (Q), (S)
(D) (II), (P), (Q)

48. Which of the following is the only CORRECT combination?
(A) (IV), (R), (T)
(B) (IV), (P), (S)
(C) (III), (P), (U)
(D) (II), (R), (T)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $A_1, A_2, ..., A_n$ be the vertices of n sided regular polygon having centre at origin. If $(A_1 A_2)^2 + (A_1 A_3)^2 = (A_1 A_4)^2$, then the value of n is

50. A teacher asks a puzzle to a student in a class but did not receive an answer. It is known that student fails to understand the puzzle one out of five times and the teacher fails to hear the reply one out of ten times. More over student would reply if he/she understood the puzzle. If the probability that student understood the puzzle is $\frac{a}{b}$ (a, b are coprime), then the value of b – a is

51. If the sum of all the solution(s) of the equation $\left(2\sec^2 x - \cos\frac{16}{9}x\right)\sec x - \left(\cos\frac{x}{3} + 2\tan x \sec x\right)\tan x = \sin\frac{x}{3}$; $x \in [-9\pi, 9\pi]$ is $k\pi$, then k is equal to

52. Let f be a real valued function defined as

$$f(x) = \cos^{-1}\left(\left\{x+\frac{1}{2}\right\} - \frac{1}{2}\right); \left[x+\frac{1}{2}\right] \text{ is even}$$

$$\sin^{-1}\left(\frac{1}{2} - \left\{x+\frac{1}{2}\right\}\right); \left[x+\frac{1}{2}\right] \text{ is odd}$$

(where [x], {x} represents greatest integer function and fractional part function respectively).

If $\int_{-\frac{1}{2}}^{\frac{23}{2}} f(x)dx = 2k\pi$, then the value of k is

53. A circle having centre C(a, b) lying on the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ cuts the rectangular hyperbola xy = 2 in four points (x_i, y_i); i = 1, 2, 3, 4. If the locus of the centroid of triangle formed by any three points of intersection is $\left(x - \frac{2a}{3}\right)\left(y - \frac{2b}{3}\right) = k^2$, then the value of $\frac{2}{k^2}$ is

54. Let ABC be a triangle right angled at C, BC = 6. Also incircle of triangle touches side AC at D. If area of trapezium PDCB is 8 square units (P being the incentre of ΔABC), then the radius of the circle touching the incircle of triangle ΔABC and passing through the vertex B and C is _____